

A Three-State Counterexample: Infinitesimal Optimality Does Not Imply Global Markovian-Coupling Optimality

A full counterexample to Remark 2.20 of arXiv:2509.23086

August 11, 2026

Abstract

Kang and Lim ask whether a coupling generator that minimizes the instantaneous cost derivative at every state must minimize the expected cost at every later time. We give a counterexample on the three-point discrete space. Both marginal chains have deterministic unit-rate jumps, the cost has nonnegative integer entries, and the instantaneous optimizer is unique at all nine joint states. Nevertheless, at time $t = 2$ it is strictly worse than a coupling that differs at only one state. Uniformization gives the exact gap $2e^{-4}(e^2 - 5) > 0$.

Status. full counterexample; likely valid. Human review is recommended.

1 The source question

Let $\mathcal{A}, \mathcal{B}$ be Feller generators on a state space Π , let $c : \Pi^2 \rightarrow [0, \infty)$, and write $\Gamma(\mathcal{A}, \mathcal{B})$ for their Feller coupling generators. The source calls $\mathcal{J}_*$ c -optimal when it attains, simultaneously at every (x, y) ,

$$(\mathcal{J}_*c)(x, y) = \inf_{\mathcal{J} \in \Gamma(\mathcal{A}, \mathcal{B})} (\mathcal{J}c)(x, y).$$

Global-in-time optimality is the stronger condition

$$(e^{t\mathcal{J}_*}c)(x, y) \leq (e^{t\mathcal{J}}c)(x, y) \quad (t \geq 0, (x, y) \in \Pi^2, \mathcal{J} \in \Gamma(\mathcal{A}, \mathcal{B})). \quad (1)$$

Proposition 2.19 proves that (1) implies c -optimality; Remark 2.20 requests a counterexample to the converse.

Remark 2.20. It is natural to ask whether the converse of Proposition 2.19 holds: namely, if $\mathcal{J}_*$ is a c -optimal coupling of $\mathcal{A}$ and $\mathcal{B}$, does it necessarily satisfy the global minimality condition (2.17) for all $\mathcal{J} \in \Gamma(\mathcal{A}, \mathcal{B})$? Preliminary considerations suggest that this implication is *not* valid in general, and constructing an explicit counterexample would be of independent interest.

2 Construction

Take $\Pi = \{0, 1, 2\}$ with the discrete topology, and define

$$f = (1, 2, 0), \quad g = (1, 0, 1).$$

Thus the marginal generators are

$$(\mathcal{A}\varphi)(x) = \varphi(f(x)) - \varphi(x), \quad (\mathcal{B}\psi)(y) = \psi(g(y)) - \psi(y).$$

Use the cost matrix, with rows indexed by x and columns by y ,

$$C = (c(x, y))_{x, y=0}^2 = \begin{pmatrix} 7 & 6 & 0 \\ 10 & 2 & 10 \\ 6 & 9 & 1 \end{pmatrix}. \quad (2)$$

Every coupling generator is determined at each (x, y) by one number $s_{xy} \in [0, 1]$. Its three possible transition rates are

$$q((x, y), (f(x), g(y))) = s_{xy}, \quad q((x, y), (f(x), y)) = q((x, y), (x, g(y))) = 1 - s_{xy}.$$

Indeed, the two marginal rate constraints force exactly this form. Conversely, every such choice defines a coupling generator. Since the space is finite and discrete, all these generators are Feller and c lies in all their domains.

Define $\mathcal{J}_*$ by

$$(s_{xy}^*) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}. \quad (3)$$

Thus $s = 1$ means a simultaneous jump, while $s = 0$ means that the two possible one-coordinate jumps each occur at rate one.

3 The generator is the unique instantaneous optimizer

Put

$$D_{xy} = c(f(x), g(y)) - c(f(x), y) - c(x, g(y)) + c(x, y).$$

For fixed (x, y) , the value $(\mathcal{J}c)(x, y)$ is an affine function of s_{xy} whose coefficient is D_{xy} . Direct substitution in (2) gives

$$(D_{xy}) = \begin{pmatrix} -7 & 7 & -14 \\ 11 & -11 & 16 \\ -4 & 4 & -2 \end{pmatrix}. \quad (4)$$

Every entry is nonzero. The unique minimizer is therefore $s_{xy} = 1$ when $D_{xy} < 0$ and $s_{xy} = 0$ when $D_{xy} > 0$, which is precisely (3). Hence $\mathcal{J}_*$ is the unique c -optimal coupling generator in the source's infinitesimal sense.

Now let $\mathcal{L}$ agree with $\mathcal{J}_*$ at every joint state except

$$s_{22}^{\mathcal{L}} = 0 \quad \text{instead of} \quad s_{22}^* = 1.$$

We prove that $\mathcal{L}$ beats $\mathcal{J}_*$ at $(2, 2)$ and time 2.

4 Exact failure at time two

Uniformize at rate two. Put

$$R_* = 2I + \mathcal{J}_*, \quad R_{\mathcal{L}} = 2I + \mathcal{L}.$$

Both $R_*/2$ and $R_{\mathcal{L}}/2$ are stochastic matrices. For $S = (2, 2)$ set

$$\Delta_k = (R_*^k c)(S) - (R_{\mathcal{L}}^k c)(S).$$

We record a short exact recurrence. Label

$$A = 01, \quad B = 11, \quad C = 00, \quad D = 20, \quad E = 02, \quad F = 21.$$

These six states never reach S , and the two matrices agree on them. If $X_k = (R^k c)(X)$ for one of these common states, then

$$\begin{aligned} A_{k+1} &= B_k + C_k, & B_{k+1} &= B_k + D_k, & C_{k+1} &= C_k + B_k, \\ D_{k+1} &= D_k + A_k, & E_{k+1} &= E_k + B_k, & F_{k+1} &= A_k + D_k. \end{aligned}$$

The initial values are

$$(A_0, B_0, C_0, D_0, E_0, F_0) = (6, 2, 7, 6, 0, 9).$$

Under R_* the S -row is $v_S + v_A$; under $R_{\mathcal{L}}$ it is $v_E + v_F$. Thus a direct calculation gives

$$\Delta_0 = 0, \quad \Delta_1 = -2, \quad \Delta_2 = 2.$$

For completeness, the displayed recurrences imply $A_k = C_k$ for $k \geq 1$ and

$$F_k - F_{k-1} = A_{k-1} = C_{k-1} \quad (k \geq 2).$$

Together with $E_k - E_{k-1} = B_{k-1}$, this shows that from $k = 2$ onward the two S -iterates have the same increment A_k . Hence

$$\Delta_k = 2 \quad (k \geq 2). \tag{5}$$

The uniformization series is absolutely convergent and gives

$$\begin{aligned} ((e^{2\mathcal{J}^*} - e^{2\mathcal{L}})c)(S) &= e^{-4} \sum_{k=0}^{\infty} \frac{2^k}{k!} \Delta_k \\ &= e^{-4} \left(-4 + 2 \sum_{k=2}^{\infty} \frac{2^k}{k!} \right) \\ &= 2e^{-4}(e^2 - 5) > 0. \end{aligned}$$

The last inequality follows already from the strict Taylor bound $e^2 > 1 + 2 + 2^2/2 = 5$. Thus

$$(e^{2\mathcal{J}^*}c)(2, 2) > (e^{2\mathcal{L}}c)(2, 2),$$

contradicting (1).

Theorem 1. *Infinitesimal c -optimality of a Feller coupling generator does not, in general, imply global-in-time c -optimality. This remains false even on a three-point state space, with deterministic unit-rate marginals, a nonnegative integer cost, and a unique instantaneous optimizer.*

5 Verification and scope

The accompanying script recomputes (4), enumerates all 2^9 extreme coupling policies to audit pointwise minimality, and checks the integer recurrence (5). No floating-point computation is used in the proof. The counterexample concerns a general finite-state cost and therefore does not contradict the source's positive all-time result for translation-invariant Lévy generators under quadratic Euclidean cost.

A bounded search of the exact question, its distinctive wording, the arXiv identifier, and later coupling literature found special simultaneous-optimality theorems but no matching explicit counterexample. Human mathematical and literature review is nevertheless recommended.

References

- [1] W. Y. Kang and T. S. Lim, *On Optimal Markovian Couplings of Lévy Processes*, arXiv:2509.23086 (2025), Remark 2.20. <https://arxiv.org/abs/2509.23086>
- [2] W. S. Kendall, M. B. Majka, and A. Mijatović, *Optimal Markovian coupling for finite activity Lévy processes*, Bernoulli 30 (2024), 2821–2845.